

```
import pandas as pd
import numpy as np
import sklearn
import matplotlib

print(f"Pandas version: {pd.__version__}")
print(f"NumPy version: {np.__version__}")
print(f"Scikit-learn version: {sklearn.__version__}")
print(f"Matplotlib version: {matplotlib.__version__}")

print("Setup successful!")

from sklearn.datasets import fetch_california_housing

# Load the dataset
housing = fetch_california_housing(as_frame=True)
df = housing.frame

print("Dataset shape:", df.shape)
print("Features:", list(housing.feature_names))
print("First 5 rows:")
print(df.head())

# Statistical summary
print(df.describe())

# Check for missing values
print("Missing values:", df.isnull().sum().sum())

from sklearn.model_selection import train_test_split

# Separate features (X) from target (y)
X = df.drop('MedHouseVal', axis=1)
y = df['MedHouseVal']

# Split data: 80% training, 20% testing
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
print("Training samples:", X_train.shape[0])  
print("Test samples:", X_test.shape[0])
```